FARHEEN BANU SD

LinkedIn | GitHub | Email | Phone | CodeChef | HackerRank

AI & Data Science Student | Aspiring Software Engineer | Machine Learning Engineer

ABOUT ME

Artificial Intelligence and Data Science undergraduate with a CGPA of 9.5/10, specializing in Software Engineering, Full-Stack Development, and Machine Learning. Experienced in building scalable, real-time applications using Python, Java, FastAPI, React.js, REST APIs, WebSockets, and cloud technologies. Proficient in designing AI-powered systems using Large Language Models (LLMs), LangGraph, Retrieval-Augmented Generation (RAG), and vector databases. Strong problem-solving skills with a solid foundation in Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, and Computer Networks, with a passion for developing reliable and high-performance software solutions.

EDUCATION

HIGHER SECONDARY
st.Mary's Convent Matric Higher Secondary School,Annur,Coimbatore — Percentage:93.5%

COLLEGE
B.Tech Artificial Intelligence And Data Science (III year)
KPR Institute Of Engineering And Technology (2024-2028) — CGPA:9.5, SGPA:9.48

SKILL

Core Computer Science : Data Structures & Algorithms, Object-Oriented Programming (OOP), Database Management Systems (DBMS), Operating Systems, Computer Networks, Software Engineering, System Design, Distributed Systems
Programming Languages : Python, Java, JavaScript, TypeScript, C, SQL
Frameworks & Technologies : React.js, Spring Boot, Spring Security, FastAPI, Flask, Node.js, Express.js, REST APIs, WebSockets, Tailwind CSS, Vite, HTML5, CSS3
Databases & Cloud : MongoDB, Firebase Firestore, Firebase Realtime Database, SQL, ChromaDB, Firebase Authentication, Firebase Storage, AWS, Cloud Technologies,Docker
Backend & DevOps : Apache Kafka, Docker, Git, GitHub, CI/CD, Maven, Postman
AI & Machine Learning : Machine Learning, Deep Learning, Natural Language Processing (NLP), Computer Vision, Large Language Models (LLMs), LangChain, LangGraph, Retrieval-Augmented Generation (RAG), Agentic AI, OpenCV, YOLOv8, Scikit-learn, Keras
Developer Tools : Docker, Git, GitHub, CI/CD, Postman, Jupyter Notebook, VS Code, Pytest, Power BI, Tableau, Microsoft Excel
Coding Profiles : CodeChef(Gold Badge 700+ problems), HackerRank (Data Structures, Algorithms, Graph Theory, Dynamic Programming)
Relevant CourseWork : Data Structures & Algorithms, Object-Oriented Programming, Database Management Systems, Operating Systems, Computer Networks, Software Engineering, System Design, Machine Learning, Artificial Intelligence, Computer Vision

PROJECTS

Lifeline-Core: AI Emergency Call Triage System | Python ,FastAPI , LangGraph ,ChromaDB — GITHUB
Technologies: Python, FastAPI, WebSockets, REST APIs, LangGraph, LangChain, Ollama, ChromaDB, Docker, STT, TTS, VAD

- Developed a scalable, real-time emergency call triage platform using Python, FastAPI, and WebSockets to process live caller audio streams and generate structured incident reports with low latency.
- Engineered a concurrent speech processing pipeline integrating Voice Activity Detection (VAD), Speech-to-Text (STT), and Text-to-Speech (TTS) services with fallback mechanisms to ensure uninterrupted call handling and improved system reliability.
- Built a Retrieval-Augmented Generation (RAG) workflow using LangGraph, LangChain, Ollama Qwen3, and ChromaDB to perform incident severity scoring, completeness checks, and dispatcher-ready summarization, achieving 90% classification accuracy across 5+ emergency categories.
- Designed scalable REST APIs, Docker-based deployment workflows, and real-time dispatcher dashboards to support multi-session incident management, reducing dispatcher information processing time by 60%.

Campus Hub: AI-Powered Academic Management Platform | React.js, Node.js, MongoDB
Technologies: React.js, TypeScript, Vite, Tailwind CSS, Node.js, Express.js, MongoDB, JWT, WebSockets, REST APIs, Hugging Face, OpenAI, Ollama

- Developed a full-stack academic management platform using React.js, Node.js, Express.js, and MongoDB, supporting 5 user roles through role-based access control, JWT authentication, and personalized dashboards.
- Engineered end-to-end workflows for assignment management, attendance tracking, placement coordination, alumni networking, and real-time messaging using 20+ REST APIs and WebSocket-based communication.
- Designed modular MongoDB schemas and scalable backend services to manage assignments, submissions, jobs, interviews, attendance records, and chat sessions, enabling secure multi-user collaboration and real-time analytics.
- Integrated AI-powered capabilities including a context-aware tutor chatbot, plagiarism detection, student risk scoring, and submission quality analysis using Hugging Face, OpenAI, and Ollama to enhance academic decision-making.

MediVault: Secure Healthcare Records Platform | React.js, Firebase, Google Drive API — GITHUB
Technologies: React.js, JavaScript, Firebase Firestore, Firebase Authentication, Google OAuth 2.0, Google Drive API, JSON, FHIR, Tailwind CSS

- Developed a cloud-based healthcare record management platform using React.js, implementing dual storage architectures with Firebase for real-time synchronization and Google Drive API for privacy-first, user-owned data storage.
- Implemented secure authentication and role-based access control using Firebase Authentication and Google OAuth 2.0, enabling seamless integration with cloud services and secure access to medical records.
- Built real-time CRUD workflows for managing 1,000+ patient records, supporting secure creation, retrieval, updates, deletion, and synchronization across distributed cloud storage systems.
- Designed FHIR-compliant data export pipelines for individual and bulk downloads in JSON and FHIR formats, improving interoperability, data portability, and integration with healthcare systems.
- Reduced patient record retrieval time by 50% through optimized cloud storage workflows and real-time synchronization while evaluating centralized and user-owned storage models for enhanced scalability and data ownership.

Student Placement Prediction System | Python, Streamlit, Scikit-learn, SQLite — GITHUB
Technologies: Python, Streamlit, Scikit-learn, Pandas, NumPy, SQLite, bcrypt

- Developed an end-to-end machine learning platform using Python and Streamlit to predict student placement outcomes based on 25 academic, technical, extracurricular, and behavioral features from a dataset of 100,000 student records.
- Designed secure user authentication and persistent data storage using bcrypt and SQLite, enabling prediction history tracking, personalized recommendations, and downloadable reports.
- Built data preprocessing and model evaluation pipelines to compare Logistic Regression, Decision Tree, and Random Forest classifiers using accuracy, precision, recall, F1-score, and confusion matrix metrics, selecting the best-performing model for deployment.
- Engineered an interactive web application with probability-based predictions, responsive dashboards, and real-time analytics to support data-driven career planning.

Student Performance AI Platform | Python, Streamlit, Scikit-learn, SQLite **GITHUB**

Technologies: Python, Streamlit, Scikit-learn, Pandas, NumPy, Plotly, Seaborn, SQLite

- Developed an end-to-end analytics platform using Python and Streamlit to predict student math scores based on 20+ demographic and academic features from a dataset of 1,000 student records.
- Designed a modular application architecture with secure authentication, prediction history tracking, and persistent data storage using SQLite, enabling personalized analytics and report generation.
- Built and evaluated Linear Regression, Decision Tree, and Random Forest models using feature engineering and comparative analysis, achieving an R^2 score of 0.87 with the selected model.
- Implemented interactive dashboards using Plotly and Seaborn to visualize performance trends, subject correlations, and prediction insights for data-driven academic decision-making.

Driver Drowsiness Detection System | Python, OpenCV, Computer Vision **GITHUB**

Technologies: Python, OpenCV, OpenCV Facemark (LBF), NumPy, SciPy, Eye Aspect Ratio (EAR)

- Developed a real-time computer vision system using Python and OpenCV to detect driver fatigue by analyzing facial landmarks and eye blink patterns from live webcam streams.
- Implemented an Eye Aspect Ratio (EAR)-based detection algorithm using 68-point facial landmark tracking, triggering automated alerts when eye closure persisted beyond configurable thresholds.
- Integrated Haar Cascade face detection, OpenCV LBF facial landmark models, and audio-visual notifications to enable accurate fatigue detection and continuous monitoring.
- Optimized the computer vision pipeline for low-latency inference on 640×480 video streams, enabling real-time performance without specialized hardware.

EXPERIENCE

Yaane Technologies | Full Stack Development Intern | Coimbatore, India | Dec 2025 - Jan 2026

MediVault: Secure Healthcare Records Platform | React.js, Firebase, Google Drive API

Technologies: React.js, JavaScript, Firebase Firestore, Firebase Authentication, Google OAuth 2.0, Google Drive API, JSON, FHIR, Tailwind CSS

- Developed a cloud-based healthcare record management platform using React.js, implementing dual storage architectures with Firebase for real-time synchronization and Google Drive API for privacy-first, user-owned data storage.
- Implemented secure authentication and role-based access control using Firebase Authentication and Google OAuth 2.0, enabling seamless integration with cloud services and secure management of 1,000+ patient records.
- Built real-time CRUD workflows and cloud-based data pipelines for secure creation, retrieval, updates, deletion, and synchronization of patient records across distributed storage systems.
- Designed standards-compliant data export workflows supporting individual and bulk downloads in JSON and FHIR formats to improve interoperability and data portability.
- Reduced patient record retrieval time by 50% through optimized cloud storage workflows and real-time synchronization while evaluating centralized and user-owned storage models for enhanced scalability and reliability.

BookMyWenue | Data Analyst Intern | Remote | Mar 2026 – Jun 2026

Technologies: Microsoft Excel, Power Query, Power BI

- Collected, validated, and structured data for 300+ venues across 4 cities, including mandapams, banquet halls, mini halls, and party venues, ensuring data accuracy and consistency.
- Designed data processing workflows using Excel and Power Query to clean, transform, and enrich venue datasets with geolocation, amenities, pricing, and nearby points-of-interest information.
- Developed interactive Power BI dashboards to visualize venue distribution, capacity analysis, city-wise trends, and amenity availability, enabling data-driven business decisions and improved venue discoverability.

Infosys Springboard Virtual Internship 7.0 | Java Full Stack Intern | Virtual | 8 WEEKS | Jul 2026 - sep 2026

MediSphere: AI-Powered Cognitive Health Twin Platform | Java, Spring Boot, Kafka, MongoDB

Technologies: Java 25, Spring Boot 4, Spring Security, Keycloak, MongoDB, Apache Kafka, Docker, REST APIs, Microservices, FHIR, SMART on FHIR

- Developed a cloud-native healthcare platform using Spring Boot microservices for patient management, digital health twins, vitals streaming, and consent management, enabling secure and scalable electronic health record processing.
- Engineered event-driven communication using Apache Kafka to stream real-time wearable device vitals and automatically synchronize patient Health Twins, reducing manual data updates and enabling continuous health monitoring.
- Implemented secure REST APIs with Keycloak-based OAuth2 authentication, role-based access control (RBAC), and SMART on FHIR integration, ensuring HIPAA-compliant access to patient health data.
- Built MongoDB-backed Patient 360 services integrating patient records, digital twins, consent information, and real-time vitals, providing a unified healthcare dashboard for clinicians and supporting AI-driven predictive healthcare workflows.

ADDITIONAL PROJECTS

- Real-Time Object Detection Using YOLOv8 | Computer Vision, OCR
- License Plate Recognition System | YOLOv8, OpenCV, OCR
- AI Label Reading Using OCR | Tesseract OCR, OpenCV
- Face Recognition System | Deep Learning, Computer Vision
- Heart Disease Prediction System | Machine Learning, Healthcare AI
- Diabetes Prediction System | Machine Learning, Keras
- Customer Review Sentiment Analysis | NLP, Scikit-learn
- Jewelaura – Low-Code WordPress eCommerce Website | WordPress
- OpenCV Computer Vision Suite | Python, OpenCV, YOLO, NumPy
- AI Meeting Buddy | Automation, Productivity, JSON/CSV Processing

CERTIFICATES AND ACHIEVEMENTS

- Academic Topper – First Year, B.Tech AI & Data Science, KPRIET
- Best Student Award – First Year, KPRIET
- Top 8% Rank – Algouniversity Tech Fellowship (ATF) among 250,000+ participants nationwide
- Samsung Innovation Campus – Coding & Programming
- NPTEL ELITE - Data Analytics With Python 83%
- NPTEL – Responsible & Safe AI Systems
- Infosys Springboard – Programming Using Java
- Infosys Springboard – Data Structures and Algorithms Using Java
- Infosys Springboard – Software Engineering and Agile Software Development
- Microsoft – Python Programming Fundamentals
- Cisco – Data Science Essentials with Python

LANGUAGES

- English
- Tamil
- Urdu
- Hindi(Basic)
- German (Basic)

LEADERSHIP & POSITIONS OF RESPONSIBILITY

- Placement Coordinator, KPR Institute of Engineering and Technology
- Core Member, Bureau of Indian Standards (BIS)
- Executive Member, Software Development Club
- Event Coordinator, Yuva Club

HACKATHONS

- Special Appreciation - Gameora Gaming Hackathon, Sri Shakthi Institute of Engineering and Technology (2026)
- 5th Place, SRiSHTi Sustainovate Paper Presentation Finals, PSG College of Technology (2025)
- Participant, THOORAL National-Level Hackathon, PSG College of Technology (2026)
- Participant, DevSpark National-Level Hackathon, KPRIET (2025)
- National level hackathon Code O Clock | CIT (Sep 26&27 2025)